# Ontology API 开发者指南

本文档提供了本体服务的两种使用方式：

1. HTTP REST API：供外部系统通过 HTTP 请求调用

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## 第一部分：HTTP REST API 接口文档

### API 概述

HTTP REST API 提供了标准的 HTTP 接口，供外部系统集成使用。所有接口支持 JSON 格式的请求和响应。

基础信息：

- 基础 URL：`https://api.onto.com`
- 请求格式：JSON
- 响应格式：JSON

### 全局响应结构

json

代码块

```
{
  "code": "success",//code为success，data中取结果，非success，message中有错误信息
  "message": "table is required",
  "requestId": "",
  "statusCode": 0,
  "data": {}
}
```

复杂查询参考：查询举例

### API 接口列表

#### API 1. 创建对象实例

接口：`POST http_rest_api/internal/v1/HttpCreateOntology`

请求体：

json

```
代码块
1  {
2          "ontology_object_type_code": "gjx_1",
3          "columns": ["name","age", "status"],
4          "values": [
5              ["Alice111",27, "active"],
6              ["Bob111",18,"inactive"]
7          ]
8  }
```

使用场景： 接收外部传感器上报的实时观测数据并入库；用户通过前端表单手动录入新的资产设备。

响应：

json

```
代码块
1  {
2    "code": "success",
3    "requestId": "",
4    "statusCode": 0,
5    "data": {
6      "sql": "INSERT INTO users (name, age, status) VALUES (?, ?, ?), (?, ?, ?)",
7      "args": [ "Alice111",27, "active","Bob111",18,"inactive"],
8      "rowsAffected": 2
9    }
10 }
```

---

### API 2. 删除对象实例

接口： `DELETE http_rest_api/internal/v1/HttpDeleteOntology`

路径参数：

- `id`：对象实例 ID

请求体：

json

```
代码块
1  {
```

```
        "ontology_object_type_code": "gjx_1",
        "where": {
                "op": "=",
                "left": {
                        "type": "column",
                        "name": "id"
                },
                "right": {
                        "type": "value",
                        "value": 60003
                }
        }
}
```

使用场景： 设备退役销毁；手动清除录入错误的垃圾测试数据。

响应：

json

代码块

```
{
  "code": "success",
  "requestId": "",
  "statusCode": 0,
  "data": {
    "sql": "DELETE FROM users WHERE id = ?",
    "args": [60003],
    "rowsAffected": 1
  }
}
```

---

### API 3. 更新对象实例属性

接口： `PUT http_rest_api/internal/v1/HttpUpdateOntology`

路径参数：

- `id`：对象实例 ID

请求体：

json

代码块

```
{
    "ontology_object_type_code": "gjx_1",
    "set": [{
        "column": "status",
        "value": {
            "type": "value",
            "value": "inactive"
        }
    }],
    "where": {
        "op": "=",
        "left": {
            "type": "column",
            "name": "id"
        },
        "right": {
            "type": "value",
            "value": 15
        }
    }
}
```

使用场景： 更新设备当前的电量、状态等动态参数；修改业务资产的归口管理部门或负责人信息。

响应：

json

代码块

```
{
    "code": "success",
    "requestId": "",
    "statusCode": 0,
    "data": {
        "sql": "UPDATE users SET status = ? WHERE id = ?",
        "args": ["inactive",15],
        "rowsAffected": 0
    }
}
```

---

### API 4. 查询对象实例列表

接口： `POST http_rest_api/internal/v1/HttpQueryOntology`

请求体：

json

代码块

```
{
    "ontology_object_type_code":"gjx_1",
    "select": [
        {
            "type": "column",
            "name": "id"
        },
        {
            "type": "column",
            "name": "name"
        }
    ],
    "where": {
        "op": "=",
        "left": {
            "type": "column",
            "name": "status"
        },
        "right": {
            "type": "value",
            "value": "active"
        }
    }
}
```

响应：

json

代码块

```
{
    "code": "success",
    "requestId": "",
    "statusCode": 0,
    "data": {
        "sql": "SELECT id, name FROM users WHERE status = ?",
        "args": ["active"],
        "results": [
            {
                "id": 9,
                "name": "Alice"
```

```
            },
            {
                "id": 60001,
                "name": "Alice"
            },
            {
                "id": 60002,
                "name": "Alice111"
            }
        ]
    }
}
```

使用场景： 管理后台展示某一类对象的完整列表；支持前端的分页展示需求。

---

### API 5. 批量删除对象实例

接口： `POST http_rest_api/internal/v1/HttpDeleteOntology`

(目前只支持条件删除，符合条件的数据可以支持批量，即不支持“where in (1，2，3)” 这种类型的批量删除)

请求体：

json

代码块

```
{
        "ontology_object_type_code": "gjx_1",
        "where": {
                "op": "=",
                "left": {
                        "type": "column",
                        "name": "id"
                },
                "right": {
                        "type": "value",
                        "value": 60003
                }
        }
}
```

使用场景： 对过期的历史任务数据进行一键清理；通过后台任务定时清除无效的临时对象。

响应：

json

```
代码块
1  {
2    "code": "success",
3    "requestId": "",
4    "statusCode": 0,
5    "data": {
6      "sql": "DELETE FROM users WHERE id = ?",
7      "args": [60003],
8      "rowsAffected": 1
9    }
10 }
```

---

### API 6. 创建对象实例关联

接口： `POST http_rest_api/internal/v1/HttpCreateOntologyLink`

（只支持N:N的关联方式）

请求体：

json

```
代码块
1  {
2          "ontology_object_type_code": "gjx-link-test",
3          "columns": [ "name","age", "status"],
4          "values": [
5              ["Alice",252222, "active"],
6              ["Bob",172222,"inactive"]
7          ]
8  }
```

响应：

json

```
代码块
1  {
2      "code": "success",
3      "requestId": "",
4      "statusCode": 0,
```

```
    "data": {
        "sql": "INSERT INTO users (name, age, status) VALUES (?, ?, ?), (?, ?, ?)",
        "args": ["Alice",252222,"active","Bob",172222,"inactive"],
        "rowsAffected": 2
    }
}
```

使用场景： 建立“无人机”与“所属站点”的业务绑定关系；将“维修记录”挂载到具体的“设备”上。

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### API 7. 删除对象实例关联

接口： `DELETE http_rest_api/internal/v1/HttpDeleteOntologyLink`

请求体：

json

代码块

```
{
    "ontology_object_type_code": "gjx-link-test",
    "where": {
        "op": "=",
        "left": {
            "type": "column",
            "name": "id"
        },
        "right": {
            "type": "value",
            "value": 60005
        }
    }
}
```

响应：

json

代码块

```
{
    "code": "success",
    "requestId": "",
    "statusCode": 0,
```

```
    "data": {
        "sql": "DELETE FROM users WHERE id = ?",
        "args": [60005],
        "rowsAffected": 1
    }
}
```

使用场景： 员工调岗解除与原部门的“所属”关系；更正由于操作失误建立的错误关联。

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### API 8. 复杂对象实例检索

接口： `POST http_rest_api/internal/v1/HttpQueryOntology`

（与API4相同）

请求体：

json

代码块

```
{
    "ontology_object_type_code":"gjx_1",
    "select": [
        {
            "type": "column",
            "name": "id"
        },
        {
            "type": "column",
            "name": "name"
        }
    ],
    "where": {
        "op": "=",
        "left": {
            "type": "column",
            "name": "status"
        },
        "right": {
            "type": "value",
            "value": "active"
        }
    }
}
```

使用场景： 精准检索（例如：找出所有“北京地区”且“剩余电量低于20%”的“小型无人机”）。

响应：

json

代码块

```
{
    "code": "success",
    "requestId": "",
    "statusCode": 0,
    "data": {
        "sql": "SELECT id, name FROM users WHERE status = ?",
        "args": [
            "active"
        ],
        "results": [
            {
                "id": 9,
                "name": "Alice"
            },
            {
                "id": 60001,
                "name": "Alice"
            },
            {
                "id": 60002,
                "name": "Alice111"
            }
        ]
    }
}
```

---

### API 9. 多对象实例关联查询 (Graph Traversal)

接口： `POST http_rest_api/internal/v1/HttpOntologyTraverse`

请求体：

json

代码块

```
{
    "code": "contract",
    "key": "mdp_contract_id",
```

```
    "value": "CT-2024-001",
    "depth": 3
}
```

使用场景： 拓扑追踪与关联追踪。例如：从某架无人机出发，查询其所属的编队，以及该编队中所有的其他关联目标。

响应：

(使用code+value作为全局唯一id)

json

代码块

```
{
    "code": "success",
    "requestId": "",
    "statusCode": 0,
    "data": {
        "nodes": [
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "code": "contract",
                "key": "mdp_contract_id",
                "value": "CT-2024-001"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:114",
                "code": "",
                "key": "mdp_customer_id",
                "value": "CUST-001"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:120",
                "code": "",
                "key": "mdp_deliverable_item_id",
                "value": "DEL-01"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:121",
                "code": "",
                "key": "mdp_deliverable_item_id",
                "value": "DEL-02"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:122",
                "code": "",
```

```
                "key": "mdp_deliverable_item_id",
                "value": "DEL-03"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:126",
                "code": "",
                "key": "mdp_rule_id",
                "value": "RULE-001"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:85",
                "code": "contract",
                "key": "mdp_contract_id",
                "value": "CT-2024-002"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:115",
                "code": "",
                "key": "mdp_customer_id",
                "value": "CUST-002"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:135",
                "code": "",
                "key": "mdp_contract_version_id",
                "value": "V-101-04"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:136",
                "code": "",
                "key": "mdp_contract_version_id",
                "value": "V-101-05"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:147",
                "code": "",
                "key": "mdp_user_id",
                "value": "U-004"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:127",
                "code": "",
                "key": "mdp_rule_id",
                "value": "RULE-002"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:112",
```

```
                "code": "contract",
                "key": "mdp_contract_id",
                "value": "CT-2024-005"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:118",
                "code": "",
                "key": "mdp_customer_id",
                "value": "CUST-005"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:144",
                "code": "",
                "key": "mdp_user_id",
                "value": "U-001"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:132",
                "code": "",
                "key": "mdp_contract_version_id",
                "value": "V-101-01"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:138",
                "code": "",
                "key": "mdp_review_record_id",
                "value": "REV-100101"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:139",
                "code": "",
                "key": "mdp_review_record_id",
                "value": "REV-100102"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:145",
                "code": "",
                "key": "mdp_user_id",
                "value": "U-002"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:140",
                "code": "",
                "key": "mdp_review_record_id",
                "value": "REV-100103"
            },
            {
```

```
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:146",
                "code": "",
                "key": "mdp_user_id",
                "value": "U-003"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:133",
                "code": "",
                "key": "mdp_contract_version_id",
                "value": "V-101-02"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:141",
                "code": "",
                "key": "mdp_review_record_id",
                "value": "REV-100104"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:142",
                "code": "",
                "key": "mdp_review_record_id",
                "value": "REV-100105"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:148",
                "code": "",
                "key": "mdp_user_id",
                "value": "U-005"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:134",
                "code": "",
                "key": "mdp_contract_version_id",
                "value": "V-101-03"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:113",
                "code": "contract",
                "key": "mdp_contract_id",
                "value": "CT-2024-006"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:119",
                "code": "",
                "key": "mdp_customer_id",
                "value": "CUST-006"
            },
```

```
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:123",
                "code": "",
                "key": "mdp_deliverable_item_id",
                "value": "DEL-04"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:124",
                "code": "",
                "key": "mdp_deliverable_item_id",
                "value": "DEL-05"
            },
            {
                "ElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:125",
                "code": "",
                "key": "mdp_deliverable_item_id",
                "value": "DEL-06"
            }
        ],
        "edges": [
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:114"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:120"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:121"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:122"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:126"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:85",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:126"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:85",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:115"
```

```
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:85",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:135"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:85",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:136"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:85",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:147"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:127"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:112",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:127"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:112",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:118"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:112",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:144"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:132"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:132",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:138"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:138",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:144"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:132",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:139"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:139",
```

```
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:145"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:132",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:140"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:140",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:146"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:133"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:133",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:141"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:141",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:147"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:133",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:142"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:142",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:148"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:134"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:144"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:113",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:144"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:113",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:119"
            },
            {
```

```
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:113",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:123"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:113",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:124"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:113",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:125"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:145"
            },
            {
                "StartElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:84",
                "EndElementId": "4:7b4dba65-c487-40e0-ae69-d8de9947b0e4:146"
            }
        ]
    }
}
```

---

## API 10. 聚合统计

接口： `POST http_rest_api/internal/v1/HttpQueryOntology`

请求体：

json

代码块

```
{
    "ontology_object_type_code":"Person",
    "select": [{
        "type": "agg",
        "func": "COUNT",
        "args": [{
            "type": "column",
            "name": "*"
        }],
        "alias": "count"
    }],
    "where": {
        "op": "=",
```

```
        "left": {
            "type": "column",
            "name": "city"
        },
        "right": {
            "type": "value",
            "value": "北京"
        }
    }
}
```

响应：

json

代码块

```
{
    "code": "success",
    "requestId": "",
    "statusCode": 0,
    "data": {
        "sql": "SELECT COUNT(*) AS count FROM user1 WHERE city = ?",
        "args": [  "北京"],
        "results": [
            {
                "count": 5
            }
        ]
    }
}
```

使用场景： 汇总统计（例如：按“在线状态”汇总各型号设备的分布总数；统计各区域资产的资产累计总价值）。

---

### API 11. 执行业务行为 (Action)

接口： `POST http_rest_api/internal/v1/HttpOntologyExecuteAction`

说明： Action 只能通过 HTTP API 调用，不能在函数（Function）中使用。

请求体：

json

代码块

```
{

    "ontologyModelID": 46,
    "projectId": "proj-1ueyp96h",
    "run_config": {
        "arguments": {
            "action123": {
                "arg1": "abc"
            },
            "action456": {
                "arg1": "def"
            }
        },
        "run_action_with_validate": true,
        "run_type": "action",
        "target": ["action123", "action456"]
    }
}
```

响应：

json

代码块

```
{
    "code": "success",
    "requestId": "ontology-dataset-0da6bca9-69d0-4f3b-9416-9cfc3affd31c",
    "statusCode": 0,
    "data": {
        "code": "success",
        "data": [
            {
                "associated_object_type": "",
                "associated_object_type_id": 0,
                "code": "",
                "created_at": "2026-03-13 16:00:37",
                "created_by": "",
                "description": "",
                "duration": 1980,
                "end_time": "2026-03-13 16:00:38",
                "execute_code": "",
                "execute_command":"from aimdp_ontology_operator.runner import run; run({\"action_def\":{},\"func_def\":{\"my_function666666\":{\"code\":\"my_function666666\",\"content\":\"# 请先修改函数名称\\n# 修改左侧的输入参数、输出参数，编辑区变量会自动修改\\n# 在右侧sdk开发指南中查看代码示例，编写函数逻辑\\n# 在参数值中通过控件输入数据\\n# 点击运行可以在下方日志区看到运行结果\\n\\ndef
```

```
my_function666666(arg1 : str) -\> dict: # 只读\\n    # 在此编写函数逻辑\\n    var_1 = 1.0\\n    var_2 = 1.0\\n    # 获取 'Person' 对象类型的引用\\n    Person = ObjectRef.Type(\\\"Person\\\")\\n    \\n    # 直接通过对象实例获取其对象类型\\n    Person = ObjectRef(\\\"Person\\\").object_type\\n    # 获取 'Person' 对象类型的引用\\n    Person = ObjectRef.Type(\\\"Person\\\")\\n    \\n    # 直接通过对象实例获取其对象类型\\n    Person = ObjectRef(\\\"Person\\\").object_type\\n    Person = ObjectRef(\\\"Person\\\").object_type\\n\\n    Person = ObjectRef(\\\"Person\\\").object_type\\n\\n    Person = ObjectRef(\\\"Person\\\").object_type\\n    pprint(Preson)\\n    return {\\\"var_1\\\":var_1,\\\"var_2\\\":var_2}\",\"description\":\"\",\"name\":\"测试函数运行\",\"parameters\": [{\"description\":\"arg1\",\"name\":\"arg1\",\"type\":\"String\"}],\"pk\":55,\"returns\": {\"var_1\":\"var_1\",\"var_2\":\"var_2\"},\"type\":\"function\"}},\"ontologyModelID\":46,\"projectId\":\"proj-1ueyp96h\",\"run_config\":{\"arguments\": {\"action123\":{\"arg1\":\"abc\"},\"action456\": {\"arg1\":\"def\"}},\"run_action_with_validate\":1,\"run_type\":\"action\",\"target\":[\"action123\",\"action456\"]}})",
                "id": 598,
                "input_params": "[]",
                "name": "",
                "operator_time": "2026-03-13 16:00:36",
                "pk": 0,
                "return_params": "null",
                "run_log": "Code generation failed: action定义 action123 未找到",
                "run_status": 3,
                "script_path": "",
                "session_id": "",
                "source": "行为测试",
                "start_time": "2026-03-13 16:00:36",
                "type": "",
                "updated_at": "2026-03-13 16:00:38",
                "updated_by": ""
            },
            {
                "associated_object_type": "",
                "associated_object_type_id": 0,
                "code": "",
                "created_at": "2026-03-13 16:00:37",
                "created_by": "",
                "description": "",
                "duration": 1980,
                "end_time": "2026-03-13 16:00:38",
                "execute_code": "",
                "execute_command":"from aimdp_ontology_operator.runner import run; run({\"action_def\":{},\"func_def\":{\"my_function666666\":
```

```
{\"code\":\"my_function666666\",\"content\":\"# 请先修改函数名称\\n# 修改左侧的输入参数、输出参数，编辑区变量会自动修改\\n# 在右侧sdk开发指南中查看代码示例，编写函数逻辑\\n# 在参数值中通过控件输入数据\\n# 点击运行可以在下方日志区看到运行结果\\n\\ndef my_function666666(arg1 : str) -\> dict: # 只读\\n    # 在此编写函数逻辑\\n    var_1 = 1.0\\n    var_2 = 1.0\\n    # 获取 'Person' 对象类型的引用\\n    Person = ObjectRef.Type(\\\"Person\\\")\\n    \\n    # 直接通过对象实例获取其对象类型\\n    Person = ObjectRef(\\\"Person\\\").object_type\\n    # 获取 'Person' 对象类型的引用\\n    Person = ObjectRef.Type(\\\"Person\\\")\\n    \\n    # 直接通过对象实例获取其对象类型\\n    Person = ObjectRef(\\\"Person\\\").object_type\\n    Person = ObjectRef(\\\"Person\\\").object_type\\n\\n    Person = ObjectRef(\\\"Person\\\").object_type\\n\\n    Person = ObjectRef(\\\"Person\\\").object_type\\n    pprint(Preson)\\n    return {\\\"var_1\\\":var_1,\\\"var_2\\\":var_2}\",\"description\":\"\",\"name\":\"测试函数运行\",\"parameters\":[{\"description\":\"arg1\",\"name\":\"arg1\",\"type\":\"String\"}],\"pk\":55,\"returns\":{\"var_1\":\"var_1\",\"var_2\":\"var_2\"},\"type\":\"function\"}},\"ontologyModelID\":46,\"projectId\":\"proj-1ueyp96h\",\"run_config\":{\"arguments\":{\"action123\":{\"arg1\":\"abc\"},\"action456\":{\"arg1\":\"def\"}},\"run_action_with_validate\":1,\"run_type\":\"action\",\"target\":[\"action123\",\"action456\"]}})",
                "id": 599,
                "input_params": "[]",
                "name": "",
                "operator_time": "2026-03-13 16:00:36",
                "pk": 0,
                "return_params": "null",
                "run_log": "Code generation failed: action定义 action123 未找到",
                "run_status": 3,
                "script_path": "",
                "session_id": "",
                "source": "行为测试",
                "start_time": "2026-03-13 16:00:36",
                "type": "",
                "updated_at": "2026-03-13 16:00:38",
                "updated_by": ""
            }
        ],
        "requestId": "",
        "statusCode": 0
    }
}
```


使用场景： 触发外部系统逻辑。例如：由大模型确认指令后，调用 API 触发无人机“立即返航”的硬件控制逻辑。

### API 12. 查询对象类型元数据 (Metadata Query)

接口： POST `/http_rest_api/internal/v1/HttpOntologyObjecTypetMeta`

说明： 获取指定对象类型的完整定义（Schema），包括属性列表和关联关系。常用于为大模型（LLM）提供建模上下文。

请求体：（三个参数，任意组合，必须传入一个）

json

代码块

```
{
    "code": "string",
    "id": 0,
    "ontologyTableName": "string"
}
```

响应：

json

代码块

```
{
    "code": "success",
    "requestId": "",
    "statusCode": 0,
    "data": {
        "code": "SUCCESS",
        "data": {
            "code": "RisktPoin",
            "createTime": "2026-03-10T03:06:39Z",
            "createUser": "",
            "description": "",
            "filePath": "",
            "icon": "object-type-coal-mine",
            "id": 117,
            "isDeleted": 0,
            "name": "风险点",
            "ontologyDbName": "aimdp_ontology_data",
            "ontologyModelID": 44,
            "ontologyPhysicalPropertiesList": [
                {
                    "columnType": "varchar(5000)",
                    "comment": "id",
```

```
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 748,
                    "isDeleted": 0,
                    "isPrimary": 1,
                    "isUse": 1,
                    "name": "id",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "contract_id",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 749,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "contract_id",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "risk_name",
                    "createTime": "2026-03-10T03:06:39Z",
```

```
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 750,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "risk_name",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "risk_type",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 751,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "risk_type",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "risk_description",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
```

```
                    "dataSourceName": "",
                    "description": "",
                    "id": 752,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "risk_description",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "risk_level",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 753,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "risk_level",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "probability",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
```

```
                    "description": "",
                    "id": 754,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "probability",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "impact",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 755,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "impact",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "risk_basis",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
```

```
                    "id": 756,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "risk_basis",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "mitigation_measure",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 757,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "mitigation_measure",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "status",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 758,
```

```
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "status",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "identified_by",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 759,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "identified_by",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "identified_by_name",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 760,
                    "isDeleted": 0,
```

```
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "identified_by_name",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "identified_time",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 761,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "identified_time",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "is_blocking",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 762,
                    "isDeleted": 0,
                    "isPrimary": 0,
```

```
                    "isUse": 1,
                    "name": "is_blocking",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                },
                {
                    "columnType": "varchar(5000)",
                    "comment": "related_opinion_id",
                    "createTime": "2026-03-10T03:06:39Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 763,
                    "isDeleted": 0,
                    "isPrimary": 0,
                    "isUse": 1,
                    "name": "related_opinion_id",
                    "objectTypeID": 117,
                    "ontologyModelID": 44,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 0,
                    "updateTime": "2026-03-10T03:06:39Z",
                    "updateUser": ""
                }
            ],
            "ontologyTableName": "ods001_contract_demo_cr_risk_point",
            "originalDbName": "ods001_contract_demo",
            "originalTableName": "cr_risk_point",
            "sourceType": 1,
            "syncFailureReason": "",
            "syncStatus": 0,
            "syncTime": null,
            "updateTime": "2026-03-10T03:06:39Z",
            "updateUser": ""
        },
```

```
        "message": "操作成功",
        "requestId": "ontology-metadata-84cc4514-96d4-403c-93b5-cc1ee0d6745f",
        "statusCode": 0
    }
}
```


使用场景： 获取指定对象类型的完整定义（Schema），包括属性列表和关联关系。常用于为大模型（LLM）提供建模上下文。

---

### API 13. 上传附件 (Upload Attachment)

接口： `POST /file/internal/v1/UploadFile`

说明： 为特定的对象实例上传关联的附件（如图片、PDF 文档等）。

请求体 (multipart/form-data)：

- `file`：待上传的文件流

响应：

json


代码块

```
{
    "code": "success",
    "requestId": "",
    "statusCode": 0,
    "data": {
        "endPoint": "10.252.216.24:30900",
        "bucket": "ontology-dataset-service-dev",
        "path": "ontology-dataset-service-dev/20260310/f782ed2b-2870-4639-bf7e-6f7a7bc709bf/upload.bin"
    }
}
```


使用场景： 为无人机巡检记录上传现场抓拍的图片；为关键资产设备绑定 PDF 格式的电子维修手册。

---

### API 14. 查询场景/全局元数据（供大模型理解）

接口： POST `http_rest_api/internal/v1/HttpOntologyObjecTypetMeta`

说明： 按 场景 或 全部场景（全局） 粒度，按需返回本体元数据，包括对象类型、链接类型、行为类型（Action）及描述。便于大模型在调用实例级 API 前，先理解「当前有哪些对象类型、它们如何通

过链接关联、哪些对象类型上有哪些可执行行为」，用于规划查询与调用。

请求示例：

http

代码块

```
{
    "id": 1
}
```

响应：

json

代码块

```
{
    "code": "success",
    "requestId": "",
    "statusCode": 0,
    "data": {
        "code": "SUCCESS",
        "data": {
            "code": "OBJECT_TYPE_001",
            "createTime": "2026-01-31T15:27:36Z",
            "createUser": "",
            "description": "口队低史半。自月计外积研教。度电以速除。什又十火导建联更。",
            "filePath": "/path/to/file",
            "icon": "icon1",
            "id": 1,
            "isDeleted": 0,
            "name": "对象类型1",
            "ontologyDbName": "aimdp_ontology_data",
            "ontologyModelID": 1,
            "ontologyPhysicalPropertiesList": [
                {
                    "columnType": "varchar(255)",
                    "comment": "属性注释",
                    "createTime": "2026-01-31T15:27:36Z",
                    "createUser": "",
                    "dataSourceName": "",
                    "description": "",
                    "id": 3,
                    "isDeleted": 0,
                    "isPrimary": 0,
```

```
                    "isUse": 0,
                    "name": "物理属性1",
                    "objectTypeID": 1,
                    "ontologyModelID": 4,
                    "ontologyObjectTypeIcon": "",
                    "ontologyObjectTypeId": 0,
                    "ontologyObjectTypeName": "",
                    "ontologyPublicPropertiesId": 0,
                    "ontologyPublicPropertiesName": "",
                    "publicPropertyID": 1,
                    "updateTime": "2026-02-01T07:00:00Z",
                    "updateUser": ""
                }
            ],
            "ontologyTableName": "original_db_original_table",
            "originalDbName": "original_db",
            "originalTableName": "original_table",
            "sourceType": 1,
            "syncFailureReason": "",
            "syncStatus": 1,
            "syncTime": "2026-02-11T01:31:26+08:00",
            "updateTime": "2026-02-27T10:49:39Z",
            "updateUser": ""
        },
        "message": "操作成功",
        "requestId": "ontology-metadata-219c43e2-602f-47c9-99e7-b5f5c6ae7577",
        "statusCode": 0
    }
}
```


---

### API 15 查询场景库列表

接口： `POST http_rest_api/internal/v1/HttpListOntologyModel`

请求体：

json

代码块

```
{
    "filter": "",//按名称和描述模糊搜索
    "order": "desc",
    "orderBy": "name",
    "pageNo": 1,
    "pageSize": 10,
```

```
    "projectID": "proj-1ueyp96h"//项目id
}
```

响应：

json

代码块

```
{
    "code": "success",
    "requestId": "ontology-dataset-9b673e83-87aa-42aa-af17-751d5e46d4bb",
    "statusCode": 0,
    "data": {
        "code": "SUCCESS",
        "data": {
            "result": [
                {
                    "createTime": "2026-01-31T04:57:47Z",
                    "createUser": "",
                    "description": "this is a description 003this is a description 003",
                    "icon": "intelligence-analysis-scene",
                    "id": 4,
                    "isDeleted": 0,
                    "name": "本体场景别删除呀",
                    "ontologyActionCounts": 2,
                    "ontologyFunctionCounts": 1,
                    "ontologyLinkTypeCounts": 6,
                    "ontologyObjectTypeCounts": 11,
                    "projectID": "proj-1ueyp96h",
                    "tagList": [],
                    "updateTime": "2026-03-11T05:11:54Z",
                    "updateUser": "admin"
                },
                {
                    "createTime": "2026-02-10T03:08:30Z",
                    "createUser": "",
                    "description": "体场景006描述",
                    "icon": "图标地址",
                    "id": 6,
                    "isDeleted": 0,
                    "name": "本体场景001",
                    "ontologyActionCounts": 0,
                    "ontologyFunctionCounts": 0,
                    "ontologyLinkTypeCounts": 0,
                    "ontologyObjectTypeCounts": 0,
```

```
                    "projectID": "proj-1ueyp96h",
                    "tagList": [],
                    "updateTime": "2026-02-25T02:12:01Z",
                    "updateUser": ""
                },
                {
                    "createTime": "2026-02-11T05:43:38Z",
                    "createUser": "",
                    "description": "",
                    "icon": "ontology-scene-1",
                    "id": 19,
                    "isDeleted": 0,
                    "name": "新建本体场景新建本体场景新建本体场景新建本体场景新建本体场景新建本体场景",
                    "ontologyActionCounts": 0,
                    "ontologyFunctionCounts": 0,
                    "ontologyLinkTypeCounts": 0,
                    "ontologyObjectTypeCounts": 0,
                    "projectID": "proj-1ueyp96h",
                    "tagList": [],
                    "updateTime": "2026-03-06T10:24:41Z",
                    "updateUser": "admin"
                },
                {
                    "createTime": "2026-02-10T10:13:57Z",
                    "createUser": "",
                    "description": "",
                    "icon": "ontology-scene-1",
                    "id": 13,
                    "isDeleted": 0,
                    "name": "新建本体场景",
                    "ontologyActionCounts": 0,
                    "ontologyFunctionCounts": 0,
                    "ontologyLinkTypeCounts": 0,
                    "ontologyObjectTypeCounts": 0,
                    "projectID": "proj-1ueyp96h",
                    "tagList": [],
                    "updateTime": "2026-02-10T10:13:57Z",
                    "updateUser": ""
                },
                {
                    "createTime": "2026-02-10T13:32:04Z",
                    "createUser": "",
                    "description": "",
                    "icon": "ontology-scene-1",
                    "id": 15,
                    "isDeleted": 0,
```

```
                    "name": "新建本体场景",
                    "ontologyActionCounts": 0,
                    "ontologyFunctionCounts": 0,
                    "ontologyLinkTypeCounts": 0,
                    "ontologyObjectTypeCounts": 0,
                    "projectID": "proj-1ueyp96h",
                    "tagList": [],
                    "updateTime": "2026-02-10T13:32:04Z",
                    "updateUser": ""
                },
                {
                    "createTime": "2026-02-12T08:13:00Z",
                    "createUser": "",
                    "description": "",
                    "icon": "ontology-scene-1",
                    "id": 25,
                    "isDeleted": 0,
                    "name": "新建本体场景",
                    "ontologyActionCounts": 0,
                    "ontologyFunctionCounts": 0,
                    "ontologyLinkTypeCounts": 0,
                    "ontologyObjectTypeCounts": 0,
                    "projectID": "proj-1ueyp96h",
                    "tagList": [],
                    "updateTime": "2026-02-12T08:13:00Z",
                    "updateUser": ""
                },
                {
                    "createTime": "2026-02-25T03:05:41Z",
                    "createUser": "",
                    "description": "",
                    "icon": "ontology-scene-1",
                    "id": 28,
                    "isDeleted": 0,
                    "name": "新建本体场景",
                    "ontologyActionCounts": 0,
                    "ontologyFunctionCounts": 0,
                    "ontologyLinkTypeCounts": 0,
                    "ontologyObjectTypeCounts": 0,
                    "projectID": "proj-1ueyp96h",
                    "tagList": [],
                    "updateTime": "2026-02-25T03:05:41Z",
                    "updateUser": ""
                },
                {
                    "createTime": "2026-02-25T03:05:51Z",
                    "createUser": "",
```

```
                    "description": "",
                    "icon": "ontology-scene-1",
                    "id": 29,
                    "isDeleted": 0,
                    "name": "新建本体场景",
                    "ontologyActionCounts": 0,
                    "ontologyFunctionCounts": 0,
                    "ontologyLinkTypeCounts": 0,
                    "ontologyObjectTypeCounts": 0,
                    "projectID": "proj-1ueyp96h",
                    "tagList": [],
                    "updateTime": "2026-02-25T03:05:51Z",
                    "updateUser": ""
                },
                {
                    "createTime": "2026-03-06T02:10:40Z",
                    "createUser": "admin",
                    "description": "合同审查和风险识别",
                    "icon": "operations-maintenance-scene",
                    "id": 42,
                    "isDeleted": 0,
                    "name": "合同审查",
                    "ontologyActionCounts": 0,
                    "ontologyFunctionCounts": 0,
                    "ontologyLinkTypeCounts": 0,
                    "ontologyObjectTypeCounts": 1,
                    "projectID": "proj-1ueyp96h",
                    "tagList": [],
                    "updateTime": "2026-03-06T02:10:40Z",
                    "updateUser": "admin"
                },
                {
                    "createTime": "2026-03-10T02:51:33Z",
                    "createUser": "admin",
                    "description": "合同-ls-勿删",
                    "icon": "security-defense-scene",
                    "id": 44,
                    "isDeleted": 0,
                    "name": "合同-ls-勿删",
                    "ontologyActionCounts": 0,
                    "ontologyFunctionCounts": 0,
                    "ontologyLinkTypeCounts": 11,
                    "ontologyObjectTypeCounts": 13,
                    "projectID": "proj-1ueyp96h",
                    "tagList": [],
                    "updateTime": "2026-03-10T02:51:33Z",
                    "updateUser": "admin"
```

```
                }
            ],
            "totalCount": 20
        },
        "message": "操作成功",
        "requestId": "ontology-metadata-d72e66ec-1b05-4df8-b124-1631d4630fc7",
        "statusCode": 0
    }
}
```

## API 16 多表关联查询

接口： `POST http_rest_api/internal/v1/HttpMultiQueryOntology`

### 1. 一对多查询

请求体：

json

代码块

```
{
    "from": {
        "ontology_object_type_code": "Ship",
        "alias": "s"
    },
    "joins": [
        {
            "type": "LEFT",
            "ontology_object_type_code": "RouteRecord",
            "alias": "r",
            "on": {
                "op": "=",
                "left": {
                    "type": "column",
                    "table": "s",
                    "name": "ship_id"
                },
                "right": {
                    "type": "column",
                    "table": "r",
                    "name": "ship_id"
                }
            }
        }
    ],
    "select": [
```

```
            {
                "type": "column",
                "table": "s",
                "name": "ship_id",
                "alias": "ship_id"
            },
            {
                "type": "column",
                "table": "s",
                "name": "name",
                "alias": "name"
            },
            {
                "type": "column",
                "table": "r",
                "name": "route_record_id",
                "alias": "route_record_id"
            },
            {
                "type": "column",
                "table": "r",
                "name": "lat",
                "alias": "lat"
            },
            {
                "type": "column",
                "table": "r",
                "name": "lon",
                "alias": "lon"
            }
        ],
        "limit": 20,
        "offset": 0
    }
```

响应：

代码块

```
{
    "code": "success",
    "requestId": "ontology-dataset-2a9647fb-81c8-4e5d-afe9-dbfadd5b4835",
    "statusCode": 0,
    "data": {
        "sql": "SELECT s.ship_id AS ship_id, s.name AS name, r.route_record_id
AS route_record_id, r.lat AS lat, r.lon AS lon FROM ship s LEFT JOIN
```

```
    RouteRecord r ON s.ship_id = r.ship_id LIMIT ? OFFSET ?",
        "args": [
            20,
            0
        ],
        "results": [
            {
                "lat": "25.6325",
                "lon": "1126.565556",
                "name": "“鲍迪奇”号海洋测量船（USNS Bowditch，T-AGS 62）",
                "route_record_id": "1",
                "ship_id": "T_AGS62"
            },
            {
                "lat": "27.246111",
                "lon": "121.347778",
                "name": "海军山东舰",
                "route_record_id": "re_ACSD17_01",
                "ship_id": "AC_SD17"
            },
            {
                "lat": "29.911944",
                "lon": "122.876667",
                "name": "海军山东舰",
                "route_record_id": "re_ACSD17_02",
                "ship_id": "AC_SD17"
            },
            {
                "lat": "34.507778",
                "lon": "122.464444",
                "name": "海军山东舰",
                "route_record_id": "re_ACSD17_03",
                "ship_id": "AC_SD17"
            },
            {
                "lat": "35.917778",
                "lon": "120.606389",
                "name": "海军山东舰",
                "route_record_id": "re_ACSD17_04",
                "ship_id": "AC_SD17"
            },
            {
                "lat": "35.917778",
                "lon": "120.606389",
                "name": "海军山东舰",
                "route_record_id": "re_ACSD17_05",
                "ship_id": "AC_SD17"
```

```
            },
            {
                "lat": "27.637222",
                "lon": "120.973333",
                "name": "055型导弹驱逐舰南昌舰",
                "route_record_id": "re_D055101_01",
                "ship_id": "D055_101"
            },
            {
                "lat": "27.637222",
                "lon": "120.973333",
                "name": "055型导弹驱逐舰南昌舰",
                "route_record_id": "re_D055101_02",
                "ship_id": "D055_101"
            },
            {
                "lat": "24.834167",
                "lon": "119.186667",
                "name": "055型导弹驱逐舰南昌舰",
                "route_record_id": "re_D055101_03",
                "ship_id": "D055_101"
            },
            {
                "lat": "24.834167",
                "lon": "119.186667",
                "name": "055型导弹驱逐舰南昌舰",
                "route_record_id": "re_D055101_04",
                "ship_id": "D055_101"
            },
            {
                "lat": "24.834167",
                "lon": "119.186667",
                "name": "055型导弹驱逐舰南昌舰",
                "route_record_id": "re_D055101_05",
                "ship_id": "D055_101"
            },
            {
                "lat": "25.6325",
                "lon": "126.565556",
                "name": "“约翰逊”号驱逐舰（USS Ralph Johnson，DDG-114）",
                "route_record_id": "re_DDG114_01",
                "ship_id": "DDG_114"
            },
            {
                "lat": "26.550278",
                "lon": "127.783056",
                "name": "“约翰逊”号驱逐舰（USS Ralph Johnson，DDG-114）",
```

```
                "route_record_id": "re_DDG114_02",
                "ship_id": "DDG_114"
            },
            {
                "lat": "26.550278",
                "lon": "127.783056",
                "name": "“约翰逊”号驱逐舰（USS Ralph Johnson，DDG-114）",
                "route_record_id": "re_DDG114_03",
                "ship_id": "DDG_114"
            },
            {
                "lat": "26.550278",
                "lon": "127.783056",
                "name": "“约翰逊”号驱逐舰（USS Ralph Johnson，DDG-114）",
                "route_record_id": "re_DDG114_04",
                "ship_id": "DDG_114"
            },
            {
                "lat": "26.550278",
                "lon": "127.783056",
                "name": "“约翰逊”号驱逐舰（USS Ralph Johnson，DDG-114）",
                "route_record_id": "re_DDG114_05",
                "ship_id": "DDG_114"
            },
            {
                "lat": "25.6325",
                "lon": "126.565556",
                "name": "“鲍迪奇”号海洋测量船（USNS Bowditch，T-AGS 62）",
                "route_record_id": "re_TAGS62_01",
                "ship_id": "T_AGS62"
            },
            {
                "lat": "26.550278",
                "lon": "127.783056",
                "name": "“鲍迪奇”号海洋测量船（USNS Bowditch，T-AGS 62）",
                "route_record_id": "re_TAGS62_02",
                "ship_id": "T_AGS62"
            },
            {
                "lat": "26.550278",
                "lon": "127.783056",
                "name": "“鲍迪奇”号海洋测量船（USNS Bowditch，T-AGS 62）",
                "route_record_id": "re_TAGS62_03",
                "ship_id": "T_AGS62"
            },
            {
                "lat": "26.550278",
```

```
                "lon": "127.783056",
                "name": "“鲍迪奇”号海洋测量船（USNS Bowditch，T-AGS 62）",
                "route_record_id": "re_TAGS62_04",
                "ship_id": "T_AGS62"
            }
        ]
    }
}
```

### 2. 多对多查询

请求参数：

代码块

```
{
    "from": {
        "ontology_object_type_code": "Link2ThreatAlert2RiskAssessment",
        "alias": "l",
        "code_type": "link"
    },
    "joins": [
        {
            "type": "INNER",
            "ontology_object_type_code": "ThreatAlert",
            "alias": "t",
            "on": {
                "op": "=",
                "left": {
                    "type": "column",
                    "table": "l",
                    "name": "alert_id"
                },
                "right": {
                    "type": "column",
                    "table": "t",
                    "name": "alert_id"
                }
            }
        },
        {
            "type": "INNER",
            "ontology_object_type_code": "RiskAssessment",
            "alias": "r",
            "on": {
                "op": "=",
                "left": {
```

```
                    "type": "column",
                    "table": "l",
                    "name": "risk_id"
                },
                "right": {
                    "type": "column",
                    "table": "r",
                    "name": "risk_id"
                }
            }
        }
    ],
    "select": [
        {
            "type": "column",
            "table": "t",
            "name": "alert_id"
        },
        {
            "type": "column",
            "table": "t",
            "name": "alert_type"
        },
        {
            "type": "column",
            "table": "r",
            "name": "risk_id"
        },
        {
            "type": "column",
            "table": "r",
            "name": "risk_level"
        }
    ]
    // "where": {
    //     "op": "AND",
    //     "conditions": [
    //         {
    //             "op": "=",
    //             "left": {
    //                 "type": "column",
    //                 "table": "t",
    //                 "name": "some_field"
    //             },
    //             "right": {
    //                 "type": "value",
    //                 "value": "some_value"
```

```
80      //             }
81      //         }
82      //     ]
83      // }
84  }
```

注意： 参数"code_type": "link" 代表该编码对应的是关联表， 可不传，默认为object 即对象实例。

返回结果：

代码块


```
{
    "code": "success",
    "requestId": "ontology-dataset-7f6abd61-34d7-48c1-a230-a2d086c20e89",
    "statusCode": 0,
    "data": {
        "sql": "SELECT t.alert_id, t.alert_type, r.risk_id, r.risk_level FROM Link_ThreatAlert_RiskAssessment l INNER JOIN ThreatAlert t ON l.alert_id = t.alert_id INNER JOIN RiskAssessment r ON l.risk_id = r.risk_id",
        "args": [],
        "results": [
            {
                "alert_id": "ALT_231027_001",
                "alert_type": "台海过航",
                "risk_id": "MeH_risk03",
                "risk_level": "中高等威胁事件"
            },
            {
                "alert_id": "ALT_231027_002",
                "alert_type": "军事演习",
                "risk_id": "MeH_risk04",
                "risk_level": "中高等威胁事件"
            },
            {
                "alert_id": "ALT_231027_003",
                "alert_type": "国际局势",
                "risk_id": "Med_risk03",
                "risk_level": "中等威胁事件"
            }
        ]
    }
}
```

## API 17 查询对象类型列表

接口： `POST http_rest_api/internal/v1/HttpListOntologyObjectType`

请求参数：

代码块

```
{
    // "ontologyModelID": 54,
    "pageNo":1,
    "pageSize":10
}
```

返回结果：

代码块


```
{
    "code": "success",
    "requestId": "ontology-dataset-fc6d4ab2-4406-424a-a7a8-8230d8d70032",
    "statusCode": 0,
    "data": {
        "code": "SUCCESS",
        "data": {
            "result": [
                {
                    "code": "demo01",
                    "createTime": "2026-03-26T01:42:03Z",
                    "createUser": "",
                    "description": "",
                    "filePath": "",
                    "icon": "object-type-3",
                    "id": 160,
                    "isDeleted": 0,
                    "name": "demo01",
                    "ontologyDbName": "aimdp_ontology_data",
                    "ontologyModelID": 54,
                    "ontologyPhysicalPropertiesList": null,
                    "ontologyTableName": "aimdp_dataset_table_20260320101345",
                    "originalDbName": "aimdp_dataset",
                    "originalTableName": "table_20260320101345",
                    "sourceType": 1,
                    "syncFailureReason": "",
                    "syncStatus": 2,
                    "syncTime": "2026-03-26T01:42:04Z",
                    "updateTime": "2026-03-26T01:42:03Z",
```

```
                    "updateUser": ""
                },
                {
                    "code": "objectType002",
                    "createTime": "2026-03-09T09:45:13Z",
                    "createUser": "",
                    "description": "",
                    "filePath": "ontology-metadata-entity-dev/entities/fgf1.csv",
                    "icon": "object-type-civil-aviation",
                    "id": 105,
                    "isDeleted": 0,
                    "name": "objectType002",
                    "ontologyDbName": "aimdp_ontology_data",
                    "ontologyModelID": 16,
                    "ontologyPhysicalPropertiesList": null,
                    "ontologyTableName": "fgf1",
                    "originalDbName": "",
                    "originalTableName": "",
                    "sourceType": 2,
                    "syncFailureReason": "",
                    "syncStatus": 2,
                    "syncTime": "2026-03-24T03:31:42Z",
                    "updateTime": "2026-03-24T03:31:43Z",
                    "updateUser": ""
                },
                {
                    "code": "zxxObjIns",
                    "createTime": "2026-03-05T09:23:19Z",
                    "createUser": "",
                    "description": "ZXX的对象实例对象",
                    "filePath": "ontology-metadata-entity-dev/entities/example_1111.csv",
                    "icon": "object-type-1",
                    "id": 96,
                    "isDeleted": 0,
                    "name": "ZXX的对象实例对象ZXX的对象实例对象ZXX的对象实例对象ZXX的对象实例对象ZXX的对象实例对象",
                    "ontologyDbName": "aimdp_ontology_data",
                    "ontologyModelID": 16,
                    "ontologyPhysicalPropertiesList": null,
                    "ontologyTableName": "example_1111",
                    "originalDbName": "",
                    "originalTableName": "",
                    "sourceType": 2,
                    "syncFailureReason": "",
                    "syncStatus": 2,
```

```
                    "syncTime": "2026-03-16T08:30:02Z",
                    "updateTime": "2026-03-24T01:50:46Z",
                    "updateUser": ""
                },
                {
                    "code": "ShipRoute123123",
                    "createTime": "2026-03-23T09:39:28Z",
                    "createUser": "",
                    "description": "",
                    "filePath": "",
                    "icon": "object-type-drone",
                    "id": 159,
                    "isDeleted": 0,
                    "name": "ShipRoute123123",
                    "ontologyDbName": "aimdp_ontology_data",
                    "ontologyModelID": 50,
                    "ontologyPhysicalPropertiesList": null,
                    "ontologyTableName": "ods435_contract_demo_ship_route",
                    "originalDbName": "ods435_contract_demo",
                    "originalTableName": "ship_route",
                    "sourceType": 1,
                    "syncFailureReason": "",
                    "syncStatus": 2,
                    "syncTime": "2026-03-23T09:39:29Z",
                    "updateTime": "2026-03-23T09:39:28Z",
                    "updateUser": ""
                },
                {
                    "code": "Ship123123",
                    "createTime": "2026-03-23T09:39:08Z",
                    "createUser": "",
                    "description": "",
                    "filePath": "",
                    "icon": "object-type-drone",
                    "id": 158,
                    "isDeleted": 0,
                    "name": "Ship123123",
                    "ontologyDbName": "aimdp_ontology_data",
                    "ontologyModelID": 50,
                    "ontologyPhysicalPropertiesList": null,
                    "ontologyTableName": "ods435_contract_demo_ship",
                    "originalDbName": "ods435_contract_demo",
                    "originalTableName": "ship",
                    "sourceType": 1,
                    "syncFailureReason": "",
                    "syncStatus": 2,
                    "syncTime": "2026-03-23T09:39:09Z",
```

```
                    "updateTime": "2026-03-23T09:39:08Z",
                    "updateUser": ""
                },
                {
                    "code": "ArgusBattleEventHandling",
                    "createTime": "2026-03-23T03:07:09Z",
                    "createUser": "",
                    "description": "",
                    "filePath": "",
                    "icon": "object-type-warship",
                    "id": 155,
                    "isDeleted": 0,
                    "name": "战例事件处置",
                    "ontologyDbName": "aimdp_ontology_data",
                    "ontologyModelID": 47,
                    "ontologyPhysicalPropertiesList": null,
                    "ontologyTableName":
    "ods434_argus_of_argus_battleeventhandling",
                    "originalDbName": "ods434_argus_of",
                    "originalTableName": "argus_battleeventhandling",
                    "sourceType": 1,
                    "syncFailureReason": "",
                    "syncStatus": 2,
                    "syncTime": "2026-03-23T10:03:11Z",
                    "updateTime": "2026-03-23T08:42:24Z",
                    "updateUser": ""
                },
                {
                    "code": "ArgusIsland",
                    "createTime": "2026-03-23T02:40:17Z",
                    "createUser": "",
                    "description": "",
                    "filePath": "",
                    "icon": "object-type-warship",
                    "id": 154,
                    "isDeleted": 0,
                    "name": "岛屿",
                    "ontologyDbName": "aimdp_ontology_data",
                    "ontologyModelID": 47,
                    "ontologyPhysicalPropertiesList": null,
                    "ontologyTableName": "ods001_argus_of_argus_island",
                    "originalDbName": "ods001_argus_of",
                    "originalTableName": "argus_island",
                    "sourceType": 1,
                    "syncFailureReason": "",
                    "syncStatus": 2,
                    "syncTime": "2026-03-23T02:40:20Z",
```

```
                    "updateTime": "2026-03-23T02:40:17Z",
                    "updateUser": ""
                },
                {
                    "code": "ArgusUavRecord",
                    "createTime": "2026-03-23T02:39:42Z",
                    "createUser": "",
                    "description": "",
                    "filePath": "",
                    "icon": "object-type-4",
                    "id": 153,
                    "isDeleted": 0,
                    "name": "无人机航迹记录",
                    "ontologyDbName": "aimdp_ontology_data",
                    "ontologyModelID": 47,
                    "ontologyPhysicalPropertiesList": null,
                    "ontologyTableName": "ods001_argus_of_argus_uavrecord",
                    "originalDbName": "ods001_argus_of",
                    "originalTableName": "argus_uavrecord",
                    "sourceType": 1,
                    "syncFailureReason": "",
                    "syncStatus": 2,
                    "syncTime": "2026-03-23T02:39:43Z",
                    "updateTime": "2026-03-23T02:39:42Z",
                    "updateUser": ""
                },
                {
                    "code": "ArgusUav",
                    "createTime": "2026-03-23T02:38:53Z",
                    "createUser": "",
                    "description": "",
                    "filePath": "",
                    "icon": "object-type-intelligence",
                    "id": 152,
                    "isDeleted": 0,
                    "name": "无人机",
                    "ontologyDbName": "aimdp_ontology_data",
                    "ontologyModelID": 47,
                    "ontologyPhysicalPropertiesList": null,
                    "ontologyTableName": "ods001_argus_of_argus_uav",
                    "originalDbName": "ods001_argus_of",
                    "originalTableName": "argus_uav",
                    "sourceType": 1,
                    "syncFailureReason": "",
                    "syncStatus": 2,
                    "syncTime": "2026-03-23T02:38:54Z",
                    "updateTime": "2026-03-23T02:38:53Z",
```

```
                    "updateUser": ""
                },
                {
                    "code": "ArgusContingencyPlan",
                    "createTime": "2026-03-23T02:38:18Z",
                    "createUser": "",
                    "description": "",
                    "filePath": "",
                    "icon": "object-type-camera-point",
                    "id": 151,
                    "isDeleted": 0,
                    "name": "预案",
                    "ontologyDbName": "aimdp_ontology_data",
                    "ontologyModelID": 47,
                    "ontologyPhysicalPropertiesList": null,
                    "ontologyTableName":
    "ods001_argus_of_argus_contingencyplan",
                    "originalDbName": "ods001_argus_of",
                    "originalTableName": "argus_contingencyplan",
                    "sourceType": 1,
                    "syncFailureReason": "",
                    "syncStatus": 2,
                    "syncTime": "2026-03-23T02:38:19Z",
                    "updateTime": "2026-03-23T02:38:18Z",
                    "updateUser": ""
                }
            ],
            "totalCount": 121
        },
        "message": "操作成功",
        "requestId": "ontology-metadata-195c928a-f9d0-4216-bb07-aec44b0c3d42",
        "statusCode": 0
    }
}
```


---

## HTTP 完整端到端集成示例

场景：外部资产管理系统（EAM）在同步一台新购入的“无人机”设备时，需要同时上传维护手册，并将其关联到“巡检中心”部门。

### 流程 1：创建无人机实例

`POST /api/ontology/object-instances`

json

```
代码块
1  {
2    "api_identifier": "CityOntology.UAV",
3    "display_name": "UAV-BJ-001",
4    "type": "FixedWing",
5    "buy_date": "2025-01-20"
6  }
```

*响应结果示例：* `{"id": "obj_uav_101", ...}`

### 流程 2：上传该设备的 PDF 维修手册

`POST /api/ontology/object-instances/obj_uav_101/attachments` *(Multipart Form Data)*

- `file`: `maintenance_manual.pdf`

### 流程 3：建立与所属部门的关联

`POST /api/ontology/object-instances/links`

json

```
代码块
1  {
2    "source_api_identifier": "CityOntology.UAV",
3    "target_api_identifier": "CityOntology.Department",
4    "sourceObjectKey": "obj_uav_101",
5    "targetObjectKey": "dept_inspection_center"
6  }
```

---

## 函数开发示例

### 场景1：合同图遍历，遍历某个合同相关的所有实例和关系

入参：

- contract_id（合同对象实例的主键id）

出参：

- node_count（实例相关的节点个数）
- edge_count （实例相关的边个数）
- graphStr （节点和边的详细信息，格式为json字符串）

函数定义：

代码块

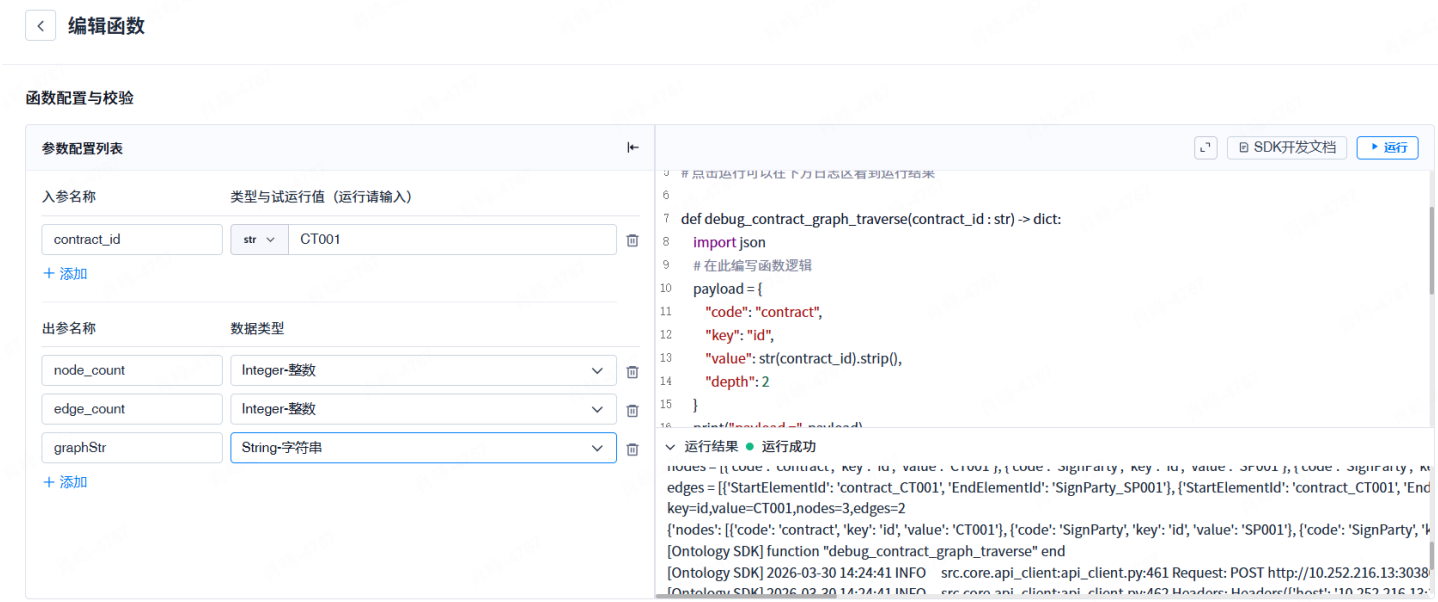
```
def debug_contract_graph_traverse(contract_id : str) -> dict:
    import json
    # 在此编写函数逻辑
    payload = {
        "code": "contract",
        "key": "id",
        "value": str(contract_id).strip(),
        "depth": 2
    }
    print("payload =", payload)
    resp = client.service.traverse_graph(payload=payload)
    print("raw resp =", resp)
    print("resp.data =", resp.data)
    data = resp.data
    nodes = data.get("nodes", [])
    edges = data.get("edges", [])
    print("nodes =", nodes)
    print("edges =", edges)
    node_count = len(nodes)
    edge_count = len(edges)
    value_str = f"key=id,value={str(contract_id).strip()},nodes=
{len(nodes)},edges={len(edges)}"
    print(value_str)
    graph = {"nodes": nodes,  "edges": edges}
    print(graph)
    graphStr = json.dumps(graph)
    return
{"node_count":node_count,"edge_count":edge_count,"graphStr":graphStr}
```

函数运行结果：

代码块

```
key=id,value=CT001,nodes=3,edges=2
{'nodes': [{'code': 'contract', 'key': 'id', 'value': 'CT001'}, {'code':
'SignParty', 'key': 'id', 'value': 'SP001'}, {'code': 'SignParty', 'key':
```

```
'id', 'value': 'SP002'}], 'edges': [{'StartElementId': 'contract_CT001',
'EndElementId': 'SignParty_SP001'}, {'StartElementId': 'contract_CT001',
'EndElementId': 'SignParty_SP002'}]}
```




## 场景2：按保密等级获取合同实例

入参：

- conf_level（保密等级）

出参：

- var_1（匹配到的合同数量）
- var_2 （匹配到的合同名称）

函数定义：

代码块

```python
def get_contracts_by_conf_level(conf_level : str) -> dict:
    print("========== 函数开始 ==========")
    print("输入的 conf_level =", conf_level)
    print("conf_level 类型 =", type(conf_level))
    level = str(conf_level).strip()
    print("标准化后的 level =", level)
    # 通过 SDK 获取 contract 对象类型
    Contract = ObjectRef.Type("contract")
    print("Contract =", Contract)
    # 使用 filter 做条件过滤
    contracts = Contract.filter(conf_level=level)
    print("contracts =", contracts)
    # 统计数量
    var_1 = contracts.count()
    print("匹配到的合同数量 var_1 =", var_1)
```

```python
    # 收集合同名称
    names = []
    for idx, contract in enumerate(contracts):
        print("第 %s 条合同对象 =" % idx, contract)
        try:
            print("contract.properties =", contract.properties)
        except Exception as e:
            print("打印 contract.properties 失败 =", e)
        try:
            name = str(contract.properties.ctt_name).strip()
        except Exception as e:
            print("获取 ctt_name 失败 =", e)
            name = ""
        print("提取出的合同名称 =", name)
        if name:
            names.append(name)
    print("合同名称列表 names =", names)
    var_2 = ",".join(names)
    print("最终返回 var_2 =", var_2)
    print("========== 函数结束 ==========")
    return {"var_1":var_1,"var_2":var_2}
```

函数运行结果：

代码块

```
合同名称列表 names = ['服务器采购合同', '智慧医院集成平台实施合同', '数据治理平台建设合同', '能源设备智能巡检服务合同', '灾备中心建设与运维合同']
```

## 使用场景对比

| 使用场景 | 推荐方式 | 说明 |
| --- | --- | --- |
| 外部系统集成 | HTTP REST API | 标准 HTTP 接口，易于集成 |
| 前端应用调用 | HTTP REST API | 通过 HTTP 请求调用 |
| 函数中编辑本体数据 | 平台开发方法 | 在函数运行时环境中使用 |
| 批量数据处理 | 平台开发方法 | 在函数中处理大量数据 |

---

## 总结